

JEE MAIN REVISION MASTER: POLYMERS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 15 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Classification of Polymers & Monomer Units	12 PYQs	Polymer Chemistry
Moderate	Biodegradable Polymers & Vulcanization	4 PYQs	Applied Polymers

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Classification of Polymers & Monomer Units	$\text{Nylon 6,6: } n \text{HOOC(CH}_2)_4\text{COOH} + n \text{H}_2\text{N(CH}_2)_6\text{NH}_2 \rightarrow n \text{[-CO(CH}_2)_4\text{CONH(CH}_2)_6\text{NH-}]_n + 2n \text{H}_2\text{O}$
Biodegradable Polymers & Vulcanization	$\text{Natural Rubber: } [\text{-CH}_2\text{-C(CH}_3)_2\text{=CH-CH}_2\text{-}]_n$ cis-isomer

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Classification of Polymers & Monomer Units (High Yield — 12 PYQs)

Core Principle: Monomers of important polymers: Nylon 6,6 from adipic acid + hexamethylenediamine. Nylon 6 from caprolactam. Terylene (Dacron) from ethylene glycol + terephthalic acid. Bakelite from phenol + formaldehyde. Buna-S from 1,3-butadiene + styrene. Neoprene from chloroprene (2-chloro-1,3-butadiene).

■ **Caution / JEE Pitfall:** Nylon 6 is made from caprolactam alone, while Nylon 6,6 is made from two different 6-carbon monomers.

■ **Speed Trick:** Thermoplastics soften on heating (polythene, PVC); thermosetting plastics undergo irreversible cross-linking (Bakelite, melamine).

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following set of polymers are used as fibre? (i) Teflon (ii) Starch (iii) Terylene (iv) Orlon..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "The biodegradable polymer obtained by polymerisation of glycine and aminocaproic acid is..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "Match the List-I with List - I in the following. List-I List-II 1. Caprolactam a. 2. Vinyl chloride b. 3. Styrene c. 4. Propene d..." [Ans: N/A]

3.2 Biodegradable Polymers & Vulcanization (Moderate Yield — 4 PYQs)

Core Principle: Biodegradable polymers: PHBV from 3-hydroxybutanoic acid + 3-hydroxypentanoic acid; Nylon 2-nylon 6 from glycine + amino caproic acid. Natural rubber is cis-1,4-polyisoprene. Vulcanization involves heating raw rubber with 5% sulphur to introduce cross-links.

■ **Caution / JEE Pitfall:** Natural rubber is the cis-isomer of 1,4-polyisoprene; the trans-isomer is Gutta-Percha.

■ **Speed Trick:** PHBV is widely used in speciality packaging and controlled drug release.

Representative Source PYQs:

- [2023 Shift] Q#3 (MCQ): "Which of the following monomers form biodegradable polymers ?..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "Which of the following monomers can undergo condensation polymerisation?..." [Ans: N/A]
- [2019 Shift] Q#1 (MCQ): "Cis-1, 4-polyisoprene is called..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Which of the following set of polymers are used as fibre? (i) Teflon (ii) Starch (iii) Terylene (iv) Orlon

- (A) (i) and (ii)
- (B) (ii) and (iii)
- (C) (iii) and (iv)
- (D) (i) and (iv)

Correct Answer: None

Solution Analysis: The polymers that are used as fibers are : Terylene : Also known as polyester, this is a synthetic polymer that is widely used as a fiber in textiles due to its durability, resistance to stretching and shrinking, and quick-drying properties. Orlon : This is a trade name for polyacrylonitrile, a synthetic polymer that is used to make acrylic fibers, known for their wool-like feel and used for sweaters, hats, and other garments. Therefore, the correct option is : Option C : (iii) and (iv)

Benchmark Problem 2: KCET 2023 (Q#3)

Which of the following monomers form biodegradable polymers ?

- (A) Ethylene glycol and phthalic acid
- (B) Caprolactum and 1, 3-butadiene
- (C) Phenol and formaldehyde
- (D) 3-hydroxybutanoic acid and 3-hydroxypentanoic acid

Correct Answer: None

Solution Analysis: PHBV is a biodegradable polymer that belongs to the family of polyhydroxyalkanoates (PHAs). It is composed of two monomeric units i.e. 3-hydroxybutanoic acid and 3-hydroxypentanoic acid.